1 2 9 0

UNIVERSIDADE D
COIMBRA

Felipe Tassari Aveiro

A Comparative Study of Post-Inference Fusion Techniques for YOLO-Based Multimodality Human Detection

Report developed within the scope of the short-term course Robotics and Applied Machine Learning.

August 2025

1. INTRODUCTION

This report was written for the fulfillment of the short-term course *Robotics and Applied Machine Learning* at the University of Coimbra. In this context, this work focuses on benchmarking the late fusion techniques of ‘Voting’ and ‘Non-Maximum Suppression’ (NMS) for human identification using three distinct sources of information. The ‘You Only Look Once’ (YOLO) object detection algorithm was employed to infer human presence in RGB, thermal, and depth images.

Moreover, the convolutional neural networks (CNN, plural: CNNs) implemented for each modality were trained on an already curated and calibrated multi-sensory dataset provided by Institute of Systems and Robotics (ISR) at the University of Coimbra. The MID-3K[1] dataset comprises 3083 scenes captured in two indoor environments, which were collected using a mobile robot equipped with the respective sensors for each modality.

In this sense, the post-inference fusion of predictions from three independently trained YOLO (v8-small) detectors is explored, following a consistent training procedure. Finally, the resulting performance in terms of mean average precision (mAP) is analyzed and a comparative evaluation of the overall perfomances is conducted.

1.1. Research Background and Contextualization

Human detection in the context of computer vision represents a fundamental capability for the perception of the environment, particularly in scenarios involving safety, navigation, or human-robot interaction (Fakrane et al., 2024; Sousa et al., 2023). While standard imagery such as RGB can be sufficient for sensing humans under ideal conditions and provide great detail, it may underperform under suboptimal settings, such as low-light environments (Mota et al., n.d., 2024; Sousa et al., 2023). To compensate for the performance degradation, multimodal approaches that incorporate complementary information from different sensors have garnered popularity in critical applications (Fakrane et al., 2024; Sousa et al., 2023).

This procedure of fusing information from sensors essentially serves to reduce the uncertainty in human presence inference, while also leveraging the unique environmental features captured by each modality, thus compensating for their individual limitations (Becker, 2023; Sousa et al., 2023). In fact, this technique enables the provision of more reliable results with improved performance (Mota et al., n.d.). In this context, the sensory modalities most commonly employed for human detection are RGB, thermal, and depth (Sousa et al., 2023), and have been extensively investigated for this purpose.

Sensor fusion techniques can be distinguished based on the stage at which the integration of information from multiple modalities occurs in the inference pipeline. In this sense, the fusion

[1] The MID-3K multi-sensory dataset and more details can be found in the GitHub repository: https://github.com/kennedyk1/MID-3K

procedure can occur at the input level by combining raw sensor data (early fusion), at the feature level by merging features from each modality extracted at specific layers (intermediate fusion), or at the decision level by aggregating the individual outputs of each model according to a defined fusion strategy (late fusion) (Mota et al., n.d.; Sousa et al., 2023). Among these, late fusion is particularly advantageous due to its flexibility in integrating pre-trained models to improve the reliability of detections (Sousa et al., 2023). Furthermore, it stands out for its ability to retain modality-specific representations while still benefiting from multi-modal reasoning.

Fakrane et al. (2024) demonstrated that better results are achieved when fusing RGB and thermal visual information using YOLOv8-small models trained on the MID-3K dataset. Similarly, Mota et al. (2024) reported superior human detection performance when applying both early and late sensor fusion techniques based on RGB, thermal, and depth modalities with this same dataset. Likewise, Sousa et al. (2023) investigated the performance improvements attained when fusing the RGB and thermal modalities using different datasets, highlighting that the best results in terms of mAP with an IoU threshold of 0.5 were obtained with late fusion strategies. It is worth noting that these findings were all achieved using YOLO series detectors.

1.2. Objectives and Methodology Overview

The previously discussed findings reinforce the relevance of exploring and comparing late fusion strategies for multi-modality human detection. The present project aims to build upon the previous body of knowledge that investigated the MID-3K dataset by evaluating two such fusion strategies in the context of YOLOv8-based human detection across RGB, thermal, and depth data modalities.

Specifically, the overarching goal is to benchmark NMS and 'Voting' to assess through performance metrics which fusion method yields superior results under a consistent setup. For this purpose, three separate YOLOv8-small models were trained independently on each data modality using the same training parameters and split configurations, which serve as a baseline for the subsequent comparison.

The primary dataset employed throughout this work and the one used for training the detectors is the MID-3K dataset. For the sake of brevity, readers are referred to Mota et al. (n.d.) for more details, as the dataset is not described in full herein. This dataset was particularly useful for this project due to the diversity of dynamic indoor scenarios it encompasses, offering more realistic perspectives of crowded environments.

The trained detectors were then validated using both unseen splits of the MID-3K dataset (one for each modality) and a subset of the AAU VAP Trimodal People Segmentation dataset (Palmero et al., 2016). The latter presents an entirely distinct scenario, with data acquired through a static configuration and using sensors with different specifications. The choice for this dataset stems from not only its availability, as trimodal datasets are not easily accessible, but also from the fact that validation of detectors in different operational contexts is essential (Fakrane et al., 2024).

Finally, a comparative evaluation was performed across the three modalities individually and their resulting fusion outputs following the two distinct fusion approaches. This process was conducted to determine the most effective approach for robust human detection.

2. RESEARCH METHODOLOGY

This project adopted the YOLO architecture as the core object detection framework due to its balance of speed, accuracy, and low computational cost. YOLO is a state-of-the-art, highly reliable, and efficient object detection algorithm based on a CNN, which simultaneously predicts bounding boxes and class probabilities directly from full images in a single evaluation, significantly reducing computational resource consumption (Ansah et al., 2025; Redmon et al., 2016; Zahoor et al., 2025). Its architecture is also designed to handle object detection across varying scales and orientations (Zahoor et al., 2025), making it particularly effective for human detection in dynamic environments.

Given these advantages, YOLO-based detectors have been extensively adopted in practical applications requiring robust and real-time human detection (Ansah et al., 2025). In this context, to ensure the required precision while maintaining low computational overhead, three separate YOLOv8 models were trained, each corresponding to a different sensor modality using the MID-3K dataset with the objective of detecting humans.

The MID-3K dataset is a readily available multimodal dataset that includes images and their corresponding labels. These were acquired with sensors that are extensively employed in human detection tasks (Sousa et al., 2023). Even though these sensors operating individually can provide satisfactory inference results, they each present inherent limitations tied to the type of data they capture and the conditions under which they operate best. RGB cameras, for instance, offer highly detailed visual information in terms of texture and color information but are extremely sensitive to lighting variations and perform poorly in low-light or occluded environments (Mota et al., 2024; Sousa et al., 2023).

Conversely, thermal cameras excel in low visibility by capturing infrared radiation, making them robust to darkness and partial occlusion; however, they lack fine-grained spatial detail and are affected by factors such as reflective surfaces or glass. Moreover, this data is particularly useful for semantic segmentation of living beings due to the emitted heat, which provides a great contrast between humans and the background (Sousa et al., 2023).

Lastly, LiDAR sensors contribute with depth information and spatial geometry, which is essential for understanding the structure of a scene (Zahoor et al., 2025), yet they may struggle with sparse data in distant regions or when representing fine details because of the inherent low-resolution representations. In this sense, each modality contributes with distinct yet complementary information about the scene. This complementarity strongly motivated the fusion strategies, as this configuration enables compensating for the weakness of individual modalities.

A standardized training pipeline was then followed for each modality, beginning with a preprocessing stage in which the dataset was split into three distinct subsets: training, validation, and test. These splits were applied separately to each modality and followed a random distribution of fixed percentages as summarized in Table 2.1.

Table 2.1: Dataset split overview for each sensor modality in MID-3K

Split	Images	Percentage of total dataset
Train	2003	~ 65%
Validation	462	~ 15%
Test	618	~ 20%

For training each individual modality, the YOLOv8 small model was adopted as the baseline detector. All models were trained from scratch, without pretrained weights, using the same configuration across modalities. Specifically, training was conducted over 50 epochs with the default hyperparameters provided by the Ultralytics framework. The input image resolution was fixed at 640×640 pixels, and early stopping was disabled by setting the patience parameter equal to the number of epochs.

The training process used the respective training and validation splits of each modality to optimize detection performance. Finally, the unseen test set was reserved for the subsequent evaluation of the best-performing models, ensuring unbiased performance assessment. On average, the entire training procedure took approximately an hour using a NVIDIA L4 GPU available in Google Colab.

To assess the performance of each trained model, two validation strategies were considered. First, the evaluation was conducted using the test split of the respective modality within the MID-3K dataset, allowing a direct measurement of the model's ability to generalize to new samples under the same domain conditions. Additionally, to further evaluate the generalization capability of each model across domains, a subset of the AAU VAP Trimodal People Segmentation dataset was utilized. Its inclusion in the validation process enabled a comparative analysis of how each modality-specific model performs when exposed to variations in acquisition conditions, scene structure, and sensor characteristics.

Following the individual evaluation of each modality, a late fusion approach was implemented to combine their predictions and potentially enhance detection performance. Two fusion strategies were considered: NMS-based fusion and voting-based fusion. In the NMS-based strategy, the bounding boxes predicted by the three models were concatenated into a unified set, along with their associated confidence scores. A global NMS operation was then applied using an IoU threshold of 0.5, retaining only the most confident boxes in overlapping regions. This step ensured that redundant detections across modalities were suppressed, resulting in a single set of filtered predictions.

Similarly, the voting-based strategy was implemented by first concatenating all bounding box predictions from the three modalities into a single set. These boxes were then compared pairwise, and overlapping boxes were grouped into clusters based on an IoU threshold of 0.5. Within each cluster, the final bounding box was computed as a confidence-weighted average of the individual boxes, and the associated detection score was set to the mean confidence of the cluster. This approach ensured that only regions consistently detected across at least two modalities were retained.

Both strategies applied post-inference using the outputs of the independently trained detectors, without modifying the internal architecture of the original YOLO models. This modular approach allowed flexible experimentation and maintained the independence of each modality-specific detector.

To further guide the fusion process, confidence scores were weighted according to the relative reliability of each modality. Specifically, thermal predictions received the highest weight, followed by RGB and then depth. These weights were applied prior to both NMS and voting, effectively biasing the final fused detections toward the most trustworthy inputs. The next section presents a comparative analysis of the detection results obtained with the standalone models and the fused outputs.

3. RESULTS & DISCUSSIONS

Standard classification and detection metrics were considered to evaluate the performance of each modality-specific detector, including mAPs and confusion matrices, offering insights into the types of prediction errors made by each model. Figure 3.1 shows the confusion matrices for the RGB-based detector. The model delivered consistently high performance across both datasets, achieving very high precision for human detection: 0.86 on MID-3K and 0.84 on AAU VAP. The RGB detector exhibited higher confusion in identifying human figures, likely due to sensitivity to lighting variations and visual clutter.

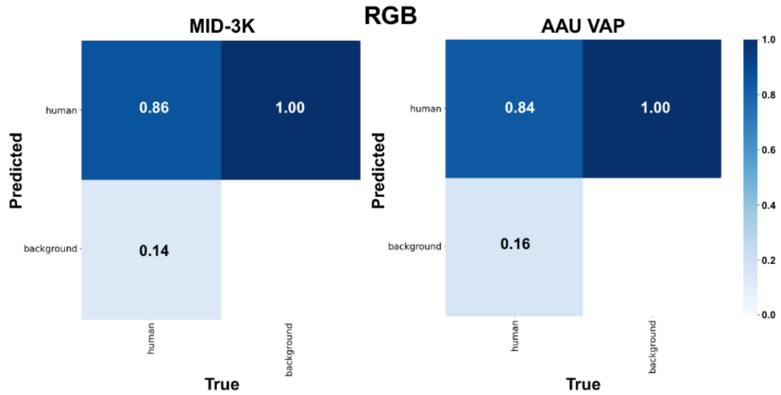

Figure 3.1: Normalized confusion matrices obtained in evaluation of the RGB detector using test splits

Figure 3.2 presents the results for the thermal-based detector. The model performed extremely well, in fact slightly higher than the RGB model. On MID-3K, only 10% of human instances were misclassified as background, while on AAU VAP, this error rate decreased to 8%. These results confirm the thermal modality's robustness, with minimal confusion and excellent discriminative power even under different environmental conditions.

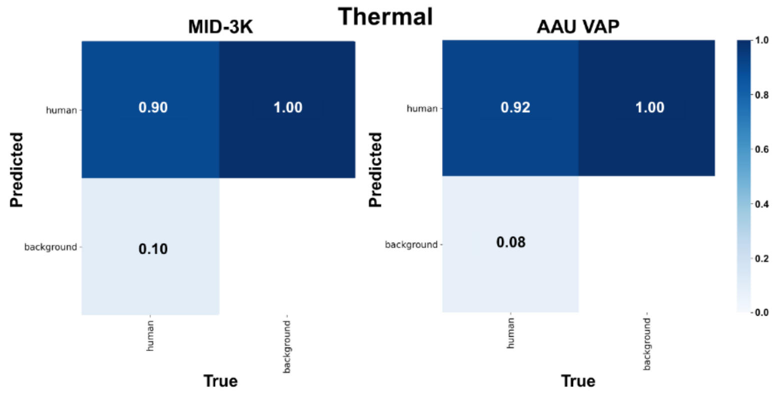

Figure 3.2: Normalized confusion matrices obtained in evaluation of the thermal detector using test splits

Figure 3.3 illustrates the depth-based detector's performance, which was markedly weaker compared to the other modalities. On MID-3K, only 65% of human instances were correctly identified, with 35% misclassified as background. On AAU VAP, the performance further degraded, with just 21% of humans correctly detected. These results reveal significant limitations in the standalone use of depth data for reliable human detection, highlighting its susceptibility to noise and poor spatial resolution in complex environments.

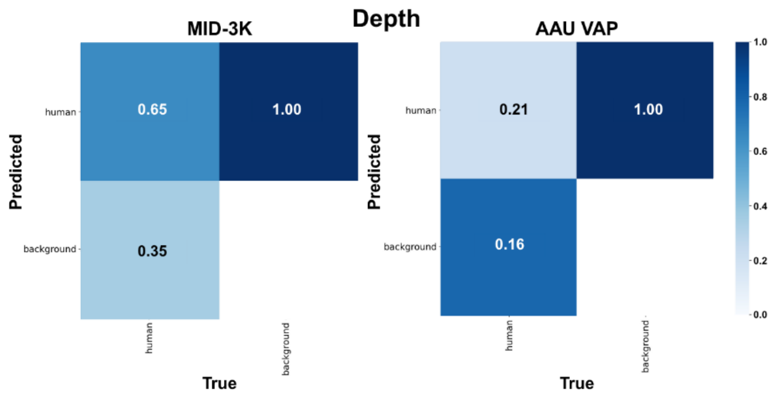

Figure 3.3: Normalized confusion matrices obtained in evaluation of the depth detector using test splits

This pattern suggests that the thermal modality provides reliable and consistent cues for human detection, even under variable backgrounds or lighting conditions. Conversely, models based on the depth modality displayed a markedly higher rate of misclassification, often confusing background regions with humans or missing actual targets.

To evaluate the effectiveness of the late fusion strategies, quantitative metrics were computed over the entire MID-3K dataset using mAP at IoU thresholds of 0.5 and the averaged range from 0.5

to 0.95 (in steps of 0.05) were used. Table 3.1 summarizes the performance of each standalone model as well as the fused results using both NMS and 'Voting' strategies.

Table 3.1: Overview of mAP@0.5 and mAP@0.5:0.95 scores

Modality	mAP@0.5	mAP@0.5:0.95
RGB (R)	0.8646	0.6399
Thermal (T)	0.8871	0.6931
Depth (D)	0.6506	0.4168
Non-Maximum Suppression (R+T+D)	0.7993	0.4838
Voting (R+T+D)	0.7714	0.5146

Contrary to common expectations, these results reveal that the thermal detector in its standalone version achieved the highest detection performance across both metrics, outperforming not only the other modalities but also the fused configurations. While fusion is commonly expected to enhance detection by integrating complementary information, the outcomes in this case suggest otherwise. Nevertheless, the NMS-based fusion strategy demonstrated superior performance in comparison to the voting-based counterpart.

Several factors may contribute to this outcome. First, the thermal modality appears to be highly informative and consistent, likely capturing strong human signatures regardless of background complexity or illumination. In contrast, the depth modality demonstrated the weakest individual performance and may have introduced noise into the fusion process. Moreover, the late fusion strategies do not learn to selectively trust more reliable predictions. As a result, predictions from weaker modalities may have been preserved during fusion, slightly degrading the final performance.

Overall, the results demonstrate the superiority of the thermal modality in standalone human detection under varied environmental conditions. Despite not surpassing the thermal-only model in terms of accuracy, the late fusion approaches still yielded satisfactory performance, confirming their practical relevance. Importantly, fusion strategies integrate complementary information from all available modalities, resulting in a more complete understanding of the scene (Fakrane et al., 2024).

This multi-perspective view is of paramount importance in safety-critical or dynamic environments, where relying on a single modality may be risky, as sensor redundancy provided by fusion increases system robustness (Sousa et al., 2023). Therefore, while fusion did not enhance accuracy in this specific setting, it remains a promising direction. Future directions should explore adaptive or learned fusion mechanisms capable of weighting modality contributions based on context or prediction confidence, potentially overcoming the limitations observed in this study.

REFERENCES

Ansah, P. A. K., Appati, J. K., Owusu, E., Boahen, E. K., Boakye-Sekyerehene, P., & Dwumfour, A. (2025). SB-YOLO-V8: A Multilayered Deep Learning Approach for Real-Time Human Detection. *Engineering Reports*, *7*(2). https://doi.org/10.1002/eng2.70033

Becker, Alex. (2023). *Kalman Filter from the Ground Up*. KilmanFilter.NET. https://books.google.pt/books/about/Kalman_Filter_from_the_Ground_Up.html?id=_2fa0AEACAAJ&redir_esc=y

Fakrane, C., Mota, K. O. S., Premebida, C., & Rocha, R. P. (2024). Human Detection Combining RGB and Thermal Modalities. *3rd IEEE International Conference on Intelligent Reality, ICIR 2024*. https://doi.org/10.1109/ICIR64558.2024.10976913

Mota, K. O. S., De Oliveira, D. S., Garrote, L., & Premebida, C. (n.d.). *Multimodal Human Detection Using YOLO and Representation Learning for Robot Perception*. https://doi.org/10.54499/UIDB/00048/2020

Mota, K. O. S., Garrote, L., & Premebida, C. (2024). Multimodal Human Detection using RGB, Thermal and LiDAR modalities for Robotic Perception. *IEEE International Conference on Automation Science and Engineering*, 4168–4173. https://doi.org/10.1109/CASE59546.2024.10711565

Palmero, C., Clapés, A., Bahnsen, C., Møgelmose, A., Moeslund, T. B., & Escalera, S. (2016). Multi-modal RGB–Depth–Thermal Human Body Segmentation. *International Journal of Computer Vision*, *118*(2), 217–239. https://doi.org/10.1007/s11263-016-0901-x

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). *You Only Look Once: Unified, Real-Time Object Detection*. http://arxiv.org/abs/1506.02640

Sousa, E., Mota, K. O. S., Gomes, I. P., Garrote, L., Wolf, D. F., & Premebida, C. (2023). Late-Fusion Multimodal Human Detection Based on RGB and Thermal Images for Robotic Perception. *Proceedings of the 11th European Conference on Mobile Robots, ECMR 2023*. https://doi.org/10.1109/ECMR59166.2023.10256301

Zahoor, L., Alhasson, H. F., Alnusayri, M., Alatiyyah, M., Alhammadi, D. A., Jalal, A., & Liu, H. (2025). Remote Sensing Surveillance Using Multilevel Feature Fusion and Deep Neural Network. *IEEE Access*, *13*, 38282–38300. https://doi.org/10.1109/ACCESS.2025.3542435